

LOG ROTATION

Real-World Industry Use Cases

BANKING / FINTECH

Scenario: A payment gateway processes 50,000 transactions/hour and logs every request.

Problem: Transaction logs grow ~5GB/day. After 30 days = 150GB on a single file.

```
/var/log/payment-gateway/*.log {
    daily
    rotate 90          # 90 days retention (PCI-DSS compliance)
    compress
    size 500M          # Also rotate if hits 500MB mid-day
    missingok
    notifempty
    dateext
    dateformat -%Y-%m-%d
    postrotate
        kill -USR1 $(cat /var/run/payment-gateway.pid)
    endscript
}
```

■ *RBI / PCI-DSS mandates transaction logs be retained for 90 days minimum.*

E-COMMERCE (FLASH SALE)

Scenario: During a sale event, Nginx handles 200k requests/minute.

Problem: Access logs explode from 1GB/day to 50GB in hours during flash sales.

```
/var/log/nginx/access.log {
    size 100M          # Rotate every 100MB regardless of time
    rotate 20
    compress
    delaycompress
    missingok
    sharedscripts
    postrotate
        nginx -s reopen
    endscript
}
```

■ *Size-based rotation keeps logs manageable during unpredictable traffic spikes.*

HEALTHCARE — HIPAA

Scenario: An EMR system logs all patient data access with strict 6-year retention.

Problem: Logs must be tamper-proof, auditable, and archived to cold storage.

```
/var/log/emr/audit.log {
  weekly
  rotate 312          # 6 years x 52 weeks
  compress
  dateext
  notifempty
  postrotate
    aws s3 cp /var/log/emr/audit.log-$(date +%Y-%m-%d).gz s3://emr-audit-archive/$(date +%Y)/
  endscript
}
```

■ Archive to S3 Glacier after rotation for long-term HIPAA-compliant cold storage.

GAMING SERVER (ANTI-CHEAT)

Scenario: Multiplayer game server logs player actions, kills, positions every second.

Problem: Logs are huge but only needed for 7 days (anti-cheat investigation window).

```
/var/log/gameserver/game.log {
  hourly              # Rotate every hour
  rotate 168          # 7 days x 24 hours = 168 files
  compress
  missingok
  size 200M
}
```

■ Each hourly file ~50MB — easy to pull logs for specific match time windows.

MICROSERVICES — DOCKER / K8S

Scenario: 30 microservices each writing logs inside containers.

Problem: No OS-level logrotate inside containers — need native Docker/K8s solutions.

```
# Docker Compose
services:
  order-service:
    logging:
      driver: "json-file"
      options:
        max-size: "50m"
        max-file: "10"

# Kubernetes sidecar
- name: log-rotator
  image: blacklabelops/logrotate
  env:
    - name: LOGROTATE_SIZE
      value: "50M"
    - name: LOGROTATE_COPIES
      value: "10"
```

■ Use Docker's built-in logging driver for containers; sidecar pattern for K8s.

SECURITY / WAF — IMMUTABLE LOGS

Scenario: A Web Application Firewall logs all blocked requests for forensic analysis.

Problem: Logs must be immutable once written — no tampering, forensic integrity required.

```
postrotate
# Hash for integrity
gzip /var/log/waf/access.log.1
sha256sum /var/log/waf/access.log.1.gz >> /var/log/waf/checksums.txt

# Make immutable (even root can't delete without chatr -i)
chatr +i /var/log/waf/access.log.1.gz
endscript
```

■ chatr +i makes the file immutable at the filesystem level — forensic-grade.

Strategy Summary

Use Case	Rotation	Retention	Special Need
Banking/Fintech	Daily + size	90 days	Compliance
E-Commerce (sale)	Size (100MB)	20 files	Spike handling
Healthcare	Weekly	6 years	Cold archive
Gaming	Hourly	7 days	Fast investigation
Microservices	Size/container	5-10 files	Docker native
Security/WAF	Daily	1 year	Immutability
Monitoring (ELK)	Daily	30 days	Filebeat integration